# Chapter 16 Notes

## Section 1

### Theorem 2.7

Here is a proof of the classical part of this theorem. Consider some model modal $\mathcal{M}$ for which the hypotheses of the problem hold

Atomic case: In this case $X$ is atomic. Thus it must be that $A = X$ and $Y = B$, meaning, because we have assumed $A \equiv B$ is valid in $\mathcal{M}$, $X \equiv Y$ (which is the same formula) is also valid in $\mathcal{M}$.

Negation case: In this case $X$ is of the form $\sim\phi$. Then either $A = X$ or $A$ is a subformula of $\phi$. In the first case, obviously it follows that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, $Y$ must be of the form $\sim\psi$ (where $\phi \equiv \psi$ is valid over $\mathcal{M}$ by assumption). Then $\phi \supset \psi$ is valid over $\mathcal{M}$ (by assumption), meaning $\sim\psi \supset \sim\phi$ is valid over $\mathcal{M}$. Similarly, $\sim\phi \supset \sim\psi$ is valid over $\mathcal{M}$ (because $\psi \supset \phi$ is valid over $\mathcal{M}$). Thus $\sim\phi \equiv \sim\psi$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

Conjunction case: In this case $X$ is of the form $\phi \wedge \psi$. Then either $A = X$ or $A$ is a subformula of either $\phi$ or $\psi$. In the first case, obviously it follwos that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, $Y$ must be of the form $\theta \wedge \gamma$, where both $\theta \equiv \phi$ and $\gamma \equiv \psi$ are true in $\mathcal{M}$. Then $(\phi \wedge \psi) \equiv (\theta \wedge \gamma)$, that is, $X \equiv Y$, being valid in $\mathcal{M}$ follows easily.

Existential case: In this case $X$ is of the form $(\exists x)\phi(x)$. Then either $A = X$ or $A$ is a subformula of $\phi(x)$. Again, the first case is obvious. In the second case, we show that (1) $(\exists x)\phi(x) \supset (\exists x)\psi(x)$ is valid over $\mathcal{M}$. So, for any world $p$ of the model, suppose that $(\exists x)\phi(x)$ is true at $p$. Then $\phi(a)$ is true at $p$ for some $a \in \mathcal{D}$. It is easy to see that this means $\psi(a)$ msut be true at $p$ (because $A \equiv B$ is assumed to be valid over $\mathcal{M}$). Thus $(\exists x)\psi(x)$ is true at $p$. Hence, because (1) is true at all possible worlds of $\mathcal{M}$, it follows that it is valid over $\mathcal{M}$. Similar reasoning shows that the converse of (1) is valid over $\mathcal{M}$. Hence $(\exists x)\phi(x) \equiv (\exists x)\psi(x)$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

### Exercise 2.1

(1) $p \vDash (X \vee Y)$ iff $p \vDash \sim(\sim X \wedge \sim Y)$ (by definition of $\vee$) iff $p \nvDash (\sim X \wedge \sim Y)$ iff either $p \nvDash \sim X$ or $p \nvDash \sim Y$ iff either $p \vDash X$ or $p \vDash Y$.

(2) $p \vDash (X \supset Y)$ iff $p \vDash \sim(X \wedge \sim Y)$ (by definition of $\supset$) iff $p \nvDash (X \wedge \sim Y)$ iff either $p \nvDash X$ or $p \nvDash \sim Y$ iff either $p \nvDash X$ or $p \vDash Y$.

(3) $p \vDash (\forall x)\phi(x)$ iff $p \vDash \sim(\exists x)\sim\phi(x)$ (by definition of $\forall$) iff $p \nvDash (\exists x)\sim\phi(x)$ iff $p \nvDash \sim\phi(d)$ for each $d \in \mathcal{D}$ iff $p \nvDash \phi(d)$ for each $d \in \mathcal{D}$.

(4) $p \vDash \Diamond X$ iff $p \vDash \sim\Box\sim X$ (by definition of $\Diamond$) iff $p \nvDash \Box\sim X$ iff for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \nvDash \sim X$ iff for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash X$.

### Exercise 2.2

(1) We first prove the validity of the $K$ axiom over every modal model. Consider some modal model $\mathcal{M}$, and one of its possible worlds $p$. We must show that $p \vDash \Box(X \supset Y) \supset (\Box X \supset \Box Y)$. It suffices to show that, if $p \vDash \Box(X \supset Y)$, then it

follows that $p \vDash (\Box X \supset \Box Y)$. So suppose $p \vDash \Box(X \supset Y)$. Then this means that, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (X \supset Y)$. It suffices to show that, if $p \vDash \Box X$, then it follows that $p \vDash \Box Y$. So suppose that $p \vDash \Box X$. Then, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash X$. But also $q \vDash (X \supset Y)$, meaning $q \vDash Y$. Thus $p \vDash \Box Y$. Hence $p \vDash (\Box X \supset \Box Y)$ holds, and consequently, $p \vDash \Box(X \supset Y) \supset (\Box \supset \Box Y)$ also holds, and thus $\Box(X \supset Y) \supset (\Box \supset \Box Y)$ is valid.

(2) We now prove the validity of the Barcan formula over every modal model. Consider some modal model $\mathcal{M}$, and one of its possible worlds $p$. We must show that $p \vDash (\forall x)\Box\phi(x) \supset \Box(\forall x)\phi(x)$. It suffices to show that, if $p \vDash (\forall x)\Box\phi(x)$, then it follows that $p \vDash \Box(\forall x)\phi(x)$. So suppose that $p \vDash (\forall x)\Box\phi(x)$. Thus $p \vDash \Box\phi(d)$ holds for each $d \in \mathcal{D}$, meaning $q \vDash \phi(d)$ holds for each $q \in \mathcal{G}$ with $p\mathcal{R}q$. To show that $p \vDash \Box(\forall x)\phi(x)$ holds, we must show that, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (\forall x)\phi(x)$. So consider some such $q$. We must show that $q \vDash \phi(d)$ holds for each $d \in \mathcal{D}$. But we have already shown this. Thus $q \vDash (\forall x)\phi(x)$, meaning $p \vDash \Box(\forall x)\phi(x)$, and consequently, the Barcan formula is valid.

## Exercise 2.3

We show that the converse of the Barcan formula, that is, $\Box(\forall x)\phi(x) \supset (\forall x)\Box\phi(x)$, is valid over every modal model. Consider any modal model $\mathcal{M}$, and one of its possible worlds $p$. It suffices to show that, if $p \vDash \Box(\forall x)\phi(x)$ holds, then $p \vDash (\forall x)\Box\phi(x)$ also holds. So suppose that $p \vDash \Box(\forall x)\phi(x)$ holds. Then, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (\forall x)\phi(x)$ holds, meaning $q \vDash \phi(d)$ holds for each $d \in \mathcal{D}$. To show that $p \vDash (\forall x)\Box\phi(x)$ holds, we must show that $p \vDash \Box\phi(d)$ holds for each $d \in \mathcal{D}$. So consider some such $d$. Then we must show that $q \vDash \phi(d)$ holds for each $q \in \mathcal{G}$ with $p\mathcal{R}q$. So consider some such $q$. But we already know that $q \vDash \phi(d)$ holds. Thus $p \vDash \Box\phi(d)$, meaning $p \vdash (\forall x)\Box\phi(x)$, meaning $p \vDash \Box(\forall x)\phi(x) \supset (\forall x)\Box\phi(x)$, and thus the converse of the Barcan formula is valid.

## Exercise 2.7

For this exercise, "Classical logic" as a reason in the derivation stands for something derived due to the assumption that the axiom system we are working in contains the standard rules for classical logic.

(a)

| | | |
|---|---|---|
| 1. | $((X \wedge Y) \supset X)$ | (Classical logic) |
| 2. | $\Box((X \wedge Y) \supset X)$ | (1, Necessitation rule) |
| 3. | $\Box(X \wedge Y) \supset \Box X$ | (2, K axiom) |

(b) For this, note that part (a) can of course be altered to prove that $\Box(X \wedge Y) \supset \Box Y$.

| | | |
|---|---|---|
| 1. | $\Box(X \wedge Y) \supset \Box X$ | (Part (a)) |
| 2. | $\Box(X \wedge Y) \supset \Box Y$ | (Part (a)) |
| 3. | $\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$ | (1, 2, Classical logic) |

(c)

| | | |
|---|---|---|
| 1. | $(X \supset (Y \supset (X \wedge Y)))$ | (Classical logic) |
| 2. | $\Box(X \supset (Y \supset (X \wedge Y)))$ | (1, Necessitation rule) |
| 3. | $\Box X \supset \Box(Y \supset (X \wedge Y))$ | (2, K axiom) |
| 4. | $\Box X \supset (\Box Y \supset \Box(X \wedge Y))$ | (3, K axiom) |
| 5. | $(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$ | (4, Classical logic) |